

Name.....

Number.....

Gosford High School



HIGHER SCHOOL CERTIFICATE

2017

TRIAL EXAMINATION

Mathematics Extension 2

- **General Instructions**
- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I

Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 13

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I 10 marks Allow approximately 15 minutes for section 1

Use the multiple-choice answer sheet for Questions 1 – 10.

- The expression for $\frac{dy}{dx}$ for the curve $x^2 - y^2 + x^3 \cos y - 6 = 0$ is
 - $\frac{-2x - 3x^2 \cos y}{2y}$
 - $\frac{2x + 3x^2 \cos y}{2y}$
 - $\frac{-2x - 3x^2 \cos y}{2y + x^3 \sin y}$
 - $\frac{2x + 3x^2 \cos y}{2y + x^3 \sin y}$
- An ellipse has equation $9x^2 + 25y^2 = 225$. The eccentricity and equation of the directrices for this ellipse are:
 - $e = \frac{4}{5}$ and $x = \pm \frac{25}{4}$
 - $e = \frac{4}{5}$ and $x = \pm 4$
 - $e = \frac{3}{5}$ and $x = \pm 4$
 - $e = \frac{3}{5}$ and $x = \pm \frac{25}{4}$
- What is the double root of the equation $x^3 - 5x^2 + 8x - 4 = 0$?
 - $x = -2$
 - $x = -1$
 - $x = 1$
 - $x = 2$

4 Given $z = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, which expression is equal to $(\bar{z})^{-1}$?

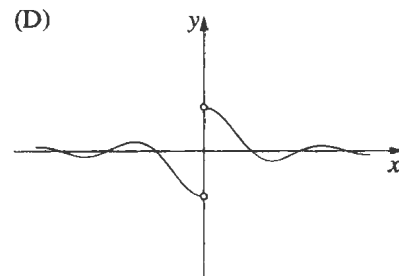
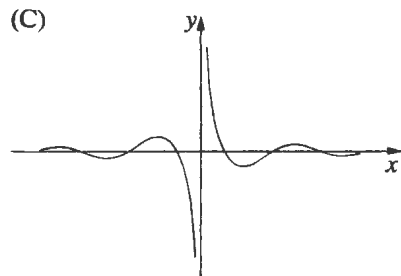
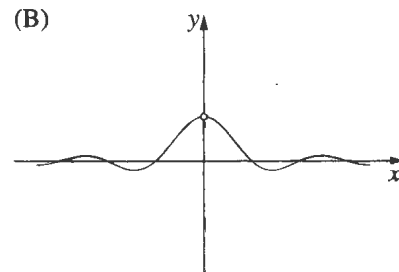
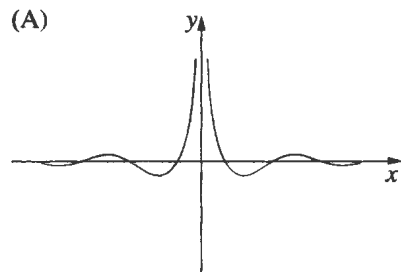
(A) $\frac{1}{2}\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

(B) $2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$

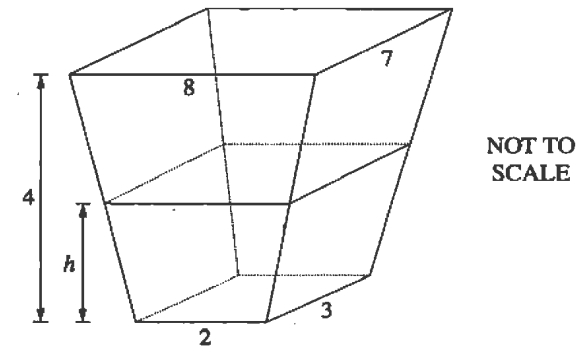
(C) $\frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

(D) $2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

5 Which diagram best represents the graph $y = \frac{\sin x}{x}$?



6 The diagram shows the dimensions of a polyhedron with parallel base and top. A slice taken at height h parallel to the base is a rectangle.



What is a correct expression for the volume of the polyhedron?

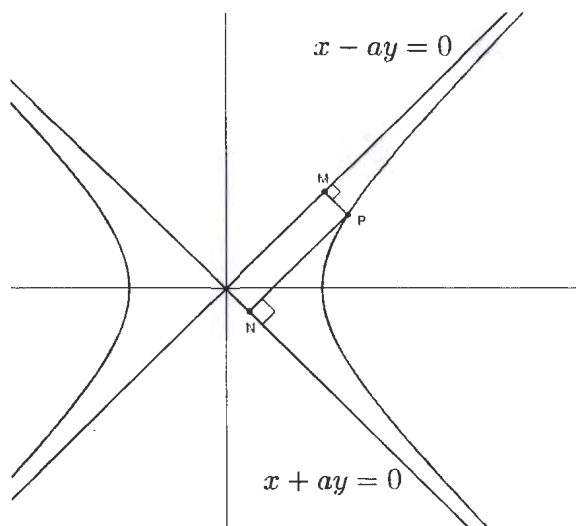
(A) $\int_0^4 (h+3)\left(\frac{3h}{2}+2\right) dh$

(B) $\int_0^4 \left(\frac{5h}{4}+3\right)\left(\frac{3h}{2}+2\right) dh$

(C) $\int_0^4 (h+3)\left(\frac{5h}{4}+2\right) dh$

(D) $\int_0^4 \left(\frac{5h}{4}+3\right)\left(\frac{5h}{4}+2\right) dh$

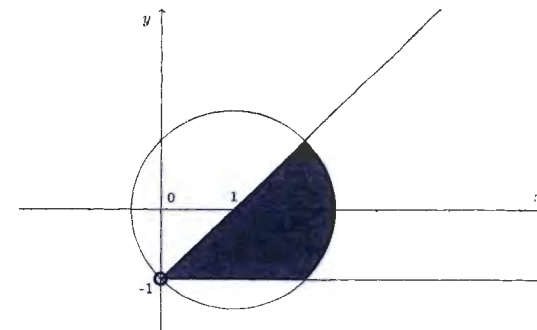
- 7) $P(a \sec \theta, \tan \theta)$ is a point on the hyperbola $\frac{x^2}{a^2} - y^2 = 1$, $a > 1$, with eccentricity e and asymptotes $x - ay = 0$ and $x + ay = 0$. M and N are the feet of the perpendiculars from P to the asymptotes as shown.



Which expression is $PM \times PN$?

- (A) $\frac{1}{e^2}$
- (B) $\frac{e^2 - 1}{e^2}$
- (C) $\frac{1}{2 - e^2}$
- (D) $\frac{1 - e^2}{2 - e^2}$

8. Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
- (B) $|z - 1| \leq \sqrt{2}$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
- (C) $|z - 1| \leq 1$ and $0 \leq \arg(z - i) \leq \frac{\pi}{4}$
- (D) $|z - 1| \leq 1$ and $0 \leq \arg(z + i) \leq \frac{\pi}{4}$
9. Consider the region bounded by the y -axis, the line $y = 4$ and the curve $y = x^2$. If this region is rotated about the line $y = 4$, which expression gives the volume of the solid of revolution?
- (A) $V = \pi \int_0^4 x^2 dy$
- (B) $V = 2\pi \int_0^2 (4 - y)x dy$
- (C) $V = \pi \int_0^2 (4 - y)^2 dx$
- (D) $V = \pi \int_0^4 (4 - y)^2 dx$
10. A committee of 5 people is to be chosen from a group of 6 girls and 4 boys. How many different committees could be formed that have at least one boy.

- (A) $\binom{10}{5} - 1$
- (B) $\binom{4}{1} + \binom{6}{4}$
- (C) $\binom{4}{1} \times \binom{6}{4}$
- (D) $\binom{10}{5} - 6$

Section II 90 marks Attempt Questions 11 – 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra booklets are available. In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Given $z = -3\sqrt{3} + 3i$
- (i) Express z in modulus-argument form 2
- (ii) Evaluate z^3 1
- (b) Use the substitution $x = 3 \tan \theta$ to find $\int \frac{dx}{x^2 \sqrt{9 + x^2}}$ 4
- (c) Find
- (i) $\int \frac{dx}{\sqrt{4 + 2x - x^2}}$ 3
- (ii) $\int \frac{2x \, dx}{x^2 + 4x + 8}$ 3
- (d) Multiplying a non-zero complex number by $\frac{1-i}{1+i}$ results in a rotation about the origin on an Argand diagram. Find the angle and direction of this rotation? 2

End of Question 11

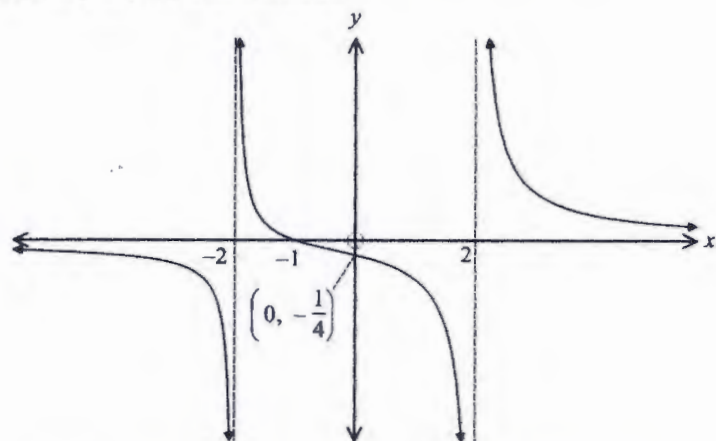
Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) The normals to the ellipse $16x^2 + 25y^2 = 400$ at the points $P(5 \cos \alpha, 4 \sin \alpha)$ and $Q(5 \cos \beta, 4 \sin \beta)$ are at right angles to each other.
- (i) Show that the gradient of the normal at P is $\frac{5 \sin \alpha}{4 \cos \alpha}$. 2
- (ii) Show that $25 \tan \alpha \tan \beta = -16$. 1
- (b) Evaluate: $\int_0^{\frac{\pi}{4}} e^x \cos 2x \, dx$ 4
- (c) (i) Find values of a , b and c such that : 3
- $$\frac{5x^2 - 3x - 8}{(x-1)(x-2)(x+2)} = \frac{a}{x-1} + \frac{b}{x-2} + \frac{c}{x+2}$$
- (ii) Hence find: $\int \frac{5x^2 - 3x - 8}{(x-1)(x-2)(x+2)} \, dx$. 1
- (d) Consider the complex numbers $\omega = -5 - 12i$ and $Z = 3 + 4i$.
- (i) Evaluate $\sqrt{\omega}$. 2
- (ii) Evaluate $\frac{\bar{\omega}}{Z}$ 2

End of Question 12

Question 13 (15 Marks) Use a SEPARATE writing booklet

- (a) The graph of $y = f(x)$ is drawn below.



Draw separate half page graphs for each of the following functions, showing all asymptotes and intercepts. **Templates are provided at the end of the paper.**

- (i) $y = |f(x)|$ 2
- (ii) $y = \frac{1}{f(x)}$ 2
- (iii) $y^2 = f(x)$ 2
- (iv) $y = e^{f(x)}$ 2
- (b) (i) Differentiate $xf(x) - \int x f'(x) dx$. 1
- (ii) Hence, or otherwise, find $\int \tan^{-1} x dx$. 2
- (c) The roots of the equation $x^3 - 9x^2 + 31x + m = 0$ are in an arithmetic sequence. Find the roots of the equation and the value of m . 4

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that a reduction formula for $I_n = \int x^n \cos x dx$ is 2
- $$I_n = x^n \sin x + nx^{n-1} \cos x - n(n-1)I_{n-2}$$
- (ii) Hence evaluate $\int x^4 \cos x dx$ 2
- (b) The area bounded by the curve $y = e^{x^2}$, the lines $x = 1$ and $y = 1$ is rotated about the y -axis. 3
- Use the method of cylindrical shells to calculate the volume of the solid of revolution formed.
- (c) The polynomial $P(x) = x^3 - 3x^2 - 4x - 5$ has roots α, β , and γ .
- (i) Find the equation with roots α^2, β^2 and γ^2 2
- (ii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$ for $P(x)$ 3
- (d) Solve $2\sin^3 \theta + 1 = 2\sin^2 \theta + \sin \theta$ for $0 \leq \theta \leq 2\pi$. 3

End of Question 14

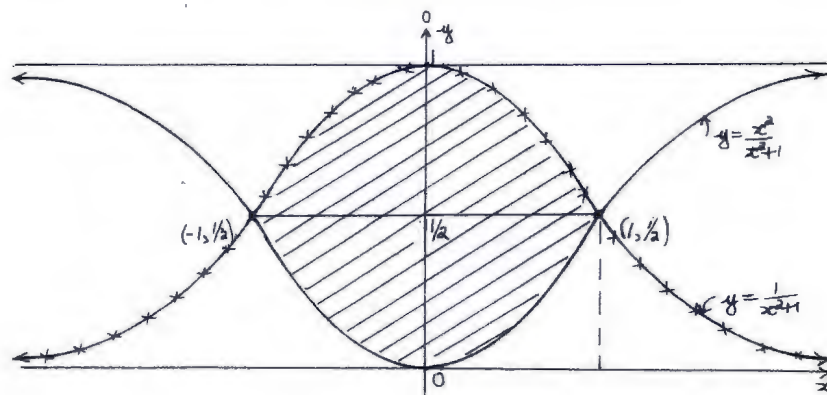
Question 15 (15 Marks) Use a SEPARATE writing booklet

- (a) Clearly sketch on an Argand diagram the locus given by

$$\arg(z-3) - \arg(z+3) = \frac{\pi}{4}$$

2

- (b)



The curves $y = \frac{1}{x^2+1}$ and $y = \frac{x^2}{x^2+1}$ are sketched above

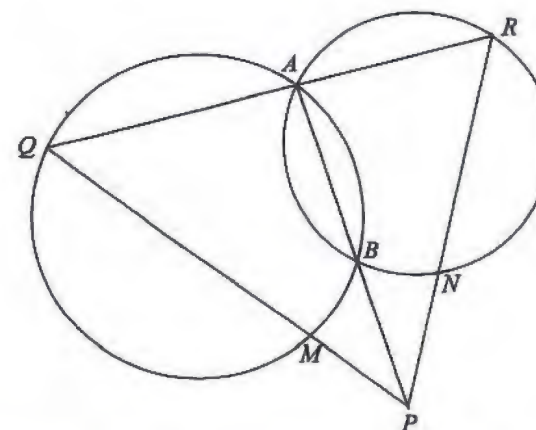
- i) Find the area bounded by the curves. 2
- ii) Find the volume of the solid generated when this area is rotated about the y-axis. 3
- c) By means of the substitution $y = a - x$ or otherwise, prove that

i) $\int_0^a f(x) dx = \int_0^a f(a-x) dx.$ 1

ii) Hence evaluate $\int_0^\pi \frac{x \sin x dx}{1+\cos^2 x}.$ 3

Question 15

- (d)



In the diagram, two circles intersect at A and B . Chord QA on the first circle is produced to cut the second circle at R . From P on AB produced secants are drawn to Q and R , cutting the circles at M and N respectively.

- (i) Show that $PMBN$ is a cyclic quadrilateral. 2
- (ii) Hence show that $MQRN$ is a cyclic quadrilateral. 2

End of Question 15

Question 16 (15 Marks) Use a SEPARATE writing booklet

- a) A particle of mass m kg is fired vertically upwards in a medium where the resistance to motion has magnitude mkv^2 newtons when the speed is v ms⁻¹. The particle has height x metres above the point of projection at time t seconds. The maximum height H metres is reached at time T seconds. The speed of projection U ms⁻¹ is equal to the terminal velocity of a particle falling in the medium. The acceleration due to gravity has magnitude g ms⁻².
- (i) Express U^2 in terms of g and k , and deduce that $\ddot{x} = -\frac{g}{U^2}(U^2 + v^2)$. 2
- (ii) Show that $\frac{v}{U} = \tan\left(\frac{\pi}{4} - \frac{g}{U}t\right)$ 2
- (iii) Show that $\frac{x}{U} = \frac{U}{g} \log\left\{\sqrt{2} \cos\left(\frac{\pi}{4} - \frac{g}{U}t\right)\right\}$ 2
- (iv) Show that at time $\frac{1}{2}T$ seconds $\frac{x}{U} = \frac{U}{2g} \log\left\{1 + \frac{1}{\sqrt{2}}\right\}$ and calculate the percentage of the maximum height attained during the first half of the ascent time, giving your answer to the nearest 1%. 3
- b) The equation of the tangent to the hyperbola $xy = c^2$ at the point $P\left(cp, \frac{c}{p}\right)$ is $x + p^2y = 2cp$. **Do not prove this.**
- (i) If the tangents at $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ meet at $R(x_0, y_0)$, Prove that $pq = \frac{x_0}{y_0}$ and $p + q = \frac{2c}{y_0}$. 2
- (ii) If the length of the chord PQ is d units, find an expression for d^2 in terms of c, p and q . (in factorised form). 2
- (iii) If d is fixed, deduce that the locus of R has equation 2
- $$4c^2(x^2 + y^2)(c^2 - xy) = x^2y^2d^2$$

End of examination

Solutions to 2017 Trial H&C Extension 2

$$x^2 - y^2 + x^3 \cos y - 6 = 0$$

$$2x - 2y \frac{dy}{dx} + x^3(-\sin y \frac{dy}{dx}) + 3x^2 \cos y = 0$$

$$2x + 3x^2 \cos y = \frac{dy}{dx} (2y + x^3 \sin y)$$

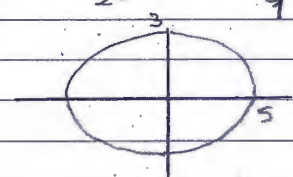
$$\frac{dy}{dx} = \frac{2x + 3x^2 \cos y}{2y + x^3 \sin y} \quad D$$

$$\frac{x^2}{25} + \frac{y^2}{9} = 1 \quad a=5, b=3$$

$$b^2 = a^2(1 - e^2)$$

$$9 = 25(1 - e^2)$$

$$\frac{9}{25} = 1 - e^2$$

$$e^2 = \frac{16}{25} \Rightarrow e = \frac{4}{5}$$


$$x = \pm \frac{a}{e} = \pm \frac{5 \times 5}{4}$$

$$x = \pm \frac{25}{4} \quad A$$

$$3x^2 - 10x + 8 = 0$$

$$(3x - 4)(x - 2) = 0$$

$$x = \frac{4}{3} \text{ or } 2 \quad D$$

$$\bar{z} = 2\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right) = 2\left(\cos -\frac{\pi}{3} + i \sin -\frac{\pi}{3}\right)$$

$$\left(\bar{z}\right)^{-1} = \frac{1}{2}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right) \quad C$$

$$\frac{\text{odd}}{\text{odd}} = \text{even} \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

B

DADC B | AABCD

$$E \quad x = m_1 h + b_1$$

$$h=0 \quad x=2$$

$$x = m_1 h + 2$$

$$h=4 \quad x=8$$

$$8 = 4m_1 + 2$$

$$\frac{3}{2} = m_1$$

$$x = \frac{3}{2}h + 2$$

$$y = m_2 h + b_2$$

$$h=0 \quad y=3$$

$$y = m_2 h + 3$$

$$h=4 \quad y=7$$

$$7 = 4m_2 + 3$$

$$m_2 = 1$$

$$y = h + 3$$

A

$$7. \quad PM, PN = \frac{(a \sec \theta - a \tan \theta)(a \sec \theta + a \tan \theta)}{1 + a^2}$$

$$= \frac{a^2(\sec^2 \theta - \tan^2 \theta)}{1 + a^2} = \frac{a^2}{1 + a^2}$$

$$b^2 = a^2(e^2 - 1)$$

$$1 = a^2(e^2 - 1)$$

$$\frac{1}{a^2} = e^2 - 1$$

$$e^2 = 1 + \frac{1}{a^2} = \frac{a^2 + 1}{a^2}$$

$$\therefore \frac{a^2}{a^2 + 1}$$

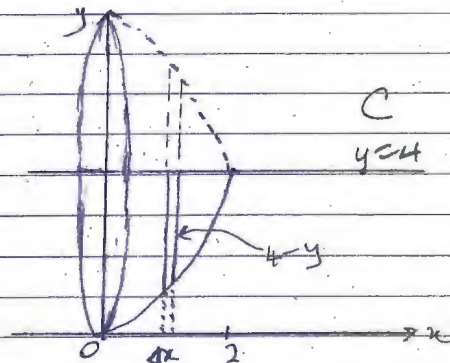
A

$$8. \quad \text{Circle } |z-1| \leq \sqrt{2}$$

$$\text{Ray } 0 \leq \arg(z+i) \leq \frac{\pi}{4}$$

B

$$9. \quad V = \pi \int_0^2 (4-y)^2 dx$$



C

y=4

$$10. \quad \text{Total - No Boys}$$

$${}^{10}C_5 - {}^6C_5 = {}^{10}C_5 - 6$$

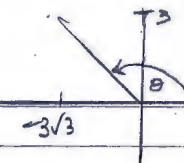
D

Question 11

$$a) i) \quad z = -3\sqrt{3} + 3i$$

$$\tan \theta = \frac{3}{-3\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$



$$|z| = \sqrt{(3\sqrt{3})^2 + 3^2} = \sqrt{27+9} = \sqrt{36} = 6$$

$$z = 6 \operatorname{cis} \frac{5\pi}{6}$$

$$ii) \quad z^3 = (6 \operatorname{cis} \frac{5\pi}{6})^3 = 6^3 \operatorname{cis} \frac{5\pi}{2}$$

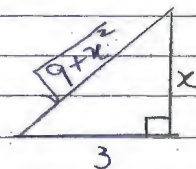
$$= 6^3 \operatorname{cis} \frac{\pi}{2} = \underline{216i}$$

$$b) \quad \int \frac{dx}{x^2 \sqrt{9+x^2}}$$

$$x = 3 \tan \theta$$

$$dx = 3 \sec^2 \theta d\theta$$

$$= \int \frac{3 \sec^2 \theta d\theta}{9 \tan^2 \theta \sqrt{9+9 \tan^2 \theta}}$$



$$= \int \frac{3 \sec^2 \theta d\theta}{9 \tan^2 \theta \cdot 3 \sec \theta}$$

$$= \frac{1}{9} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

$$= \frac{1}{9} \int \frac{1}{\cos \theta} \times \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$= \frac{1}{9} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$= \frac{1}{9} \int (\sin \theta)^{-2} \cos \theta d\theta$$

$$= \frac{1}{9} \frac{(\sin \theta)^{-1}}{-1} = -\frac{1}{9 \sin \theta}$$

$$= -\frac{\sqrt{9+x^2}}{9x} + c$$

$$c) \int \frac{dx}{\sqrt{5-(x-1)^2}} \quad 4-(x^2-2x+1)+1$$

$$= \sin^{-1} \frac{x-1}{\sqrt{5}} + c$$

$$ii) \int \frac{2x+4-4}{x^2+4x+8} dx$$

$$= \int \frac{2x+4}{x^2+4x+8} dx - \int \frac{4}{(x+2)^2+4} dx$$

$$= \ln(x^2+4x+8) - 4 \cdot \frac{1}{2} \tan^{-1} \frac{x+2}{2}$$

$$= \ln(x^2+4x+8) - 2 \tan^{-1} \frac{x+2}{2}$$

$$d) \frac{1-i}{1+i} \times \frac{1-i}{1-i} = \frac{1-2i+i^2}{1+1}$$

$$= \frac{-2i}{2} = -i$$

$\frac{\pi}{2}$ clockwise

Question 12

$$a) \frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$x = 5 \cos \theta$$

$$y = 4 \sin \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} / \frac{dx}{d\theta}$$

$$m = \frac{4 \cos \theta}{-5 \sin \theta} \text{ for tangent at P}$$

$$m_1 = \frac{5 \sin \alpha}{4 \cos \alpha} \text{ for normal at P}$$

as $m m_1 = -1$

$$ii) \text{ Normal at P } m_1 = \frac{5 \sin \alpha}{4 \cos \alpha}$$

$$\text{Normal at Q } m_2 = \frac{5 \sin \beta}{4 \cos \beta}$$

These are at right angles

$$\frac{5 \sin \alpha}{4 \cos \alpha} \times \frac{5 \sin \beta}{4 \cos \beta} = -1$$

$$25 \tan \alpha \tan \beta = -1$$

$$b) \int e^x \cos 2x dx \quad u = \cos 2x \quad dv = e^x dx$$

$$du = -2 \sin 2x dx \quad v = e^x$$

$$I = e^x \cos 2x - \int e^x (-2 \sin 2x) dx$$

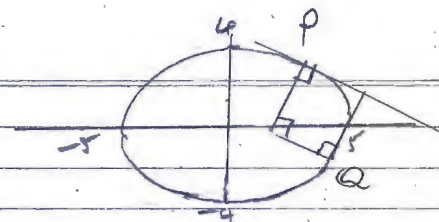
$$= e^x \cos 2x + 2 \int e^x \sin 2x dx$$

$$\begin{cases} u = \sin 2x & dv = e^x dx \\ du = 2 \cos 2x dx & v = e^x \end{cases}$$

$$I = e^x \cos 2x + 2 \left\{ e^x \sin 2x - 2 \int \cos 2x e^x dx \right\}$$

$$= e^x \cos 2x + 2e^x \sin 2x - 4 \int \cos 2x e^x dx$$

$$5I = e^x \{ \cos 2x + 2 \sin 2x \}$$



$$\int_0^{\pi/4} e^{5x} \cos 2x \, dx = \left[\frac{e^{5x}}{5} \{ \cos 2x + 2 \sin 2x \} \right]_0^{\pi/4}$$

$$= \frac{e^{\pi/4}}{5} \{ \cos \frac{\pi}{2} + 2 \sin \frac{\pi}{2} \} - \frac{e^0}{5} \{ \cos 0 + 2 \sin 0 \}$$

$$= \frac{1}{5} \{ 2e^{\pi/4} - 1 \}$$

$$c) \frac{5x^2 - 3x - 8}{(x-1)(x-2)(x+2)} = \frac{a}{x-1} + \frac{b}{x-2} + \frac{c}{x+2}$$

$$x=1 \quad 5-3-8 = -3a \Rightarrow a = -2$$

$$x=2 \quad 20-6-8 = 4b \Rightarrow b = \frac{3}{2}$$

$$x=-2 \quad 20+6-8 = (-3)(-2)c$$

$$18 = 12c \Rightarrow c = \frac{3}{2}$$

$$\int \frac{5x^2 - 3x - 8}{(x-1)(x-2)(x+2)} = \int \frac{2dx}{x-1} + \frac{3}{2} \int \frac{dx}{x-2} + \frac{3}{2} \int \frac{dx}{x+2}$$

$$= 2 \ln|x-1| + \frac{3}{2} \ln|x-2| + \frac{3}{2} \ln|x+2| + C$$

$$d) w = -5 - 12i \quad z = 3 + 4i$$

$$1) (a+ib)^2 = -5 - 12i$$

$$a+ib = \sqrt{-5-12i}$$

$$a^2 - b^2 = -5$$

$$2ab = -12$$

$$b = -\frac{6}{a}$$

$$a^2 - \frac{36}{a^2} = -5$$

$$a^4 + 5a^2 - 36 = 0$$

$$(a^2 + 9)(a^2 - 4) = 0$$

$$a = \pm 2 \quad \text{as } a \text{ is real}$$

$$b = \mp 3$$

$\therefore (2-3i)$ and $(-2+3i)$
are square roots of w

Q12

$$ii) \frac{-5+12i}{3+4i} \times \frac{3-4i}{3-4i}$$

$$\frac{-15 + 20i + 36i + 48}{9 + 16}$$

$$\frac{33 + 56i}{25}$$

Question 13 b

$$i) \frac{d}{dx} x f(x) = \int x f(x) dx$$

$$= x \cdot f(x) + f(x) \cdot 1 = x f(x)$$

$$= f(x)$$

$$ii) f(x) = \tan^{-1} x$$

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \int x \cdot \frac{1}{1+x^2} dx$$

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$$

$$c) x^3 - 9x^2 + 31x + m = 0$$

Let roots be $\alpha-d, \alpha, \alpha+d$

$$\sum \alpha = 3\alpha = 9 \Rightarrow \alpha = 3$$

Sub $\alpha = 3$

$$27 - 81 + 93 + m = 0$$

$$m = -39$$

$$\therefore x^3 - 9x^2 + 31x - 39 = 0$$

Exp:

$$3(3-d) + 3(3+d) + (3-d)(3+d) = 31$$

$$9 - 3d + 9 + 3d + 9 - d^2 = 31$$

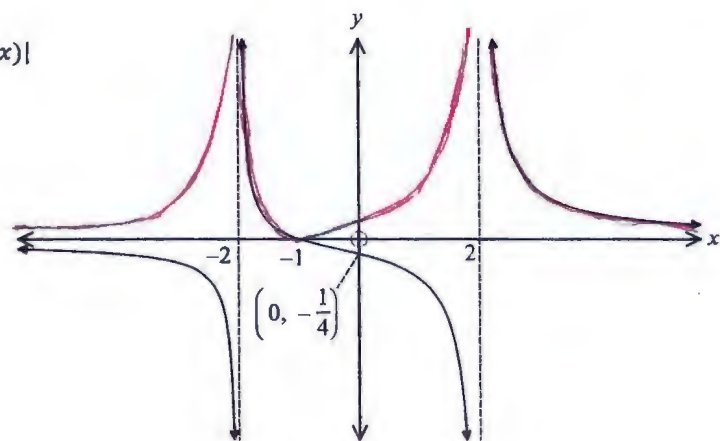
$$-4 = -d^2 \Rightarrow d = \pm 2i$$

$\therefore$ Roots are $3-2i, 3, 3+2i$

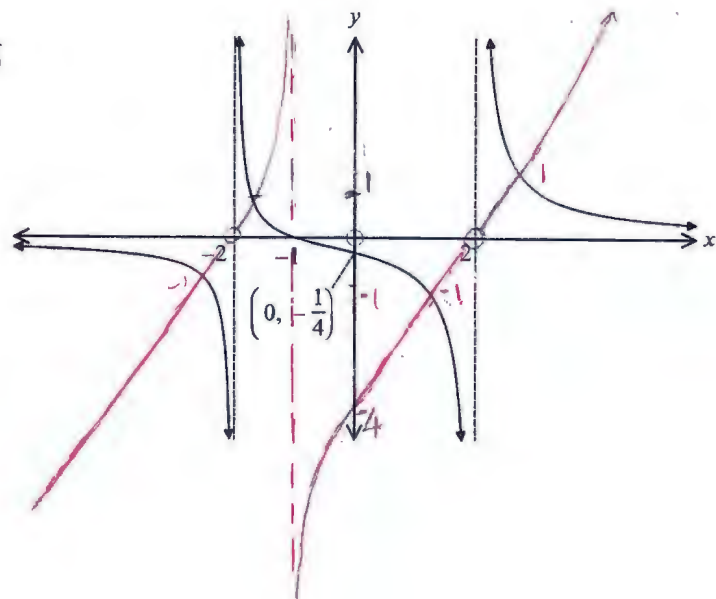
13 a)

name / number _____

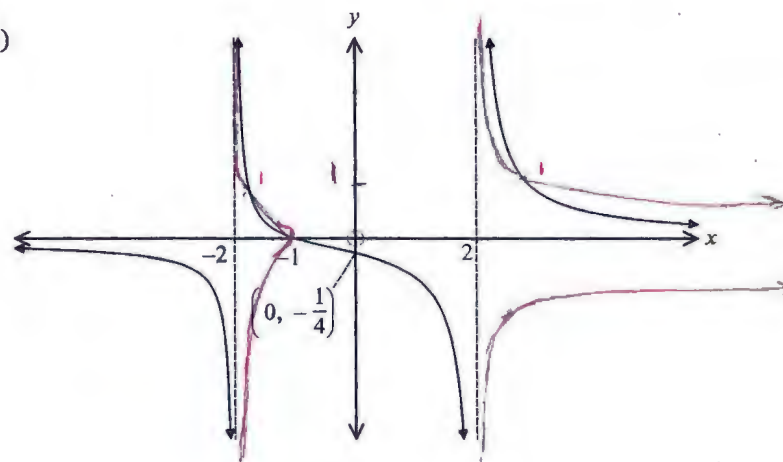
(i) $y = |f(x)|$



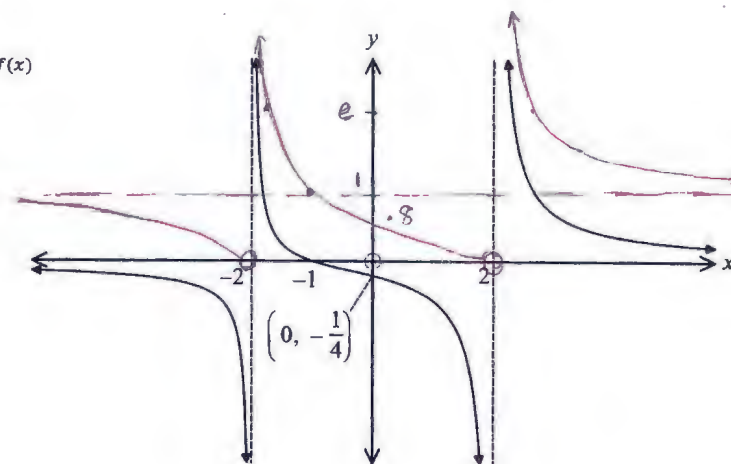
(ii) $y = \frac{1}{f(x)}$



(iii) $y^2 = f(x)$



(iv) $y = e^{f(x)}$



Question 14

a) i) $I_n = \int x^n \cos x \, dx$

$$u = x^n \quad dv = \cos x \, dx$$

$$du = n x^{n-1} \, dx \quad v = \sin x$$

$$I_n = x^n \sin x - \int \sin x \cdot n x^{n-1} \, dx$$

$$= x^n \sin x - n \int x^{n-1} \sin x \, dx$$

$$\int x^{n-1} \sin x \, dx$$

$$u = x^{n-1} \quad dv = \sin x \, dx$$

$$du = (n-1)x^{n-2} \, dx \quad v = -\cos x$$

$$= -x^{n-1} \cos x + \int \cos x (n-1)x^{n-2} \, dx$$

$$\therefore I_n = x^n \sin x - n \left\{ -x^{n-1} \cos x + (n-1) \int \cos x x^{n-2} \, dx \right\}$$

$$= x^n \sin x + n x^{n-1} \cos x - n(n-1) \int \cos x x^{n-2} \, dx$$

$$= x^n \sin x + n x^{n-1} \cos x - n(n-1) I_{n-2}$$

ii)

$$I_4 = x^4 \sin x + 4x^3 \cos x - 12 I_2$$

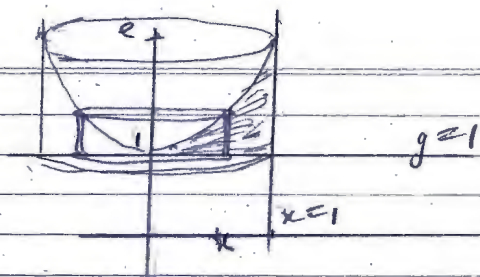
$$I_2 = x^2 \sin x + 2x \cos x - 2 I_0$$

$$I_0 = \int \cos x \, dx = \sin x$$

$$I_2 = x^2 \sin x + 2x \cos x - 2 \sin x$$

$$I_4 = x^4 \sin x + 4x^3 \cos x - 12x^2 \sin x - 24x \cos x + 24 \sin x$$

(14b)



$$V = 2\pi \int x(y-1) \, dx$$

$$= 2\pi \int_0^1 x(e^{x^2} - 1) \, dx$$

$$= 2\pi \left\{ \int_0^1 x e^{x^2} \, dx - \int_0^1 x \, dx \right\}$$

$$= 2\pi \left[\frac{1}{2} e^{x^2} - \frac{x^2}{2} \right]_0^1$$

$$= \pi (e - 1 - 1 + 0)$$

$$V = \pi (e - 2) \text{ u}^3$$

c) i) Put $y = x^2 \Rightarrow x = \pm\sqrt{y}$

$$(\pm\sqrt{y})^3 - 3(\pm\sqrt{y})^2 - 4(\pm\sqrt{y}) - 5 = 0$$

$$\pm\sqrt{y} y - 3y - 4(\pm\sqrt{y}) - 5 = 0$$

$$\pm\sqrt{y} (y - 4) = 3y + 5$$

$$y(y - 4)^2 = (3y + 5)^2$$

$$y(y^2 - 8y + 16) = 9y^2 + 30y + 25$$

$$y^3 - 8y^2 + 16y = 9y^2 + 30y + 25$$

$$y^3 - 17y^2 - 14y - 25 = 0$$

Note $\sum x^2 = 11$

14 c)
ii)

$$P(x) = x^3 - 3x^2 - 4x - 5$$

$$P(\alpha) = \alpha^3 - 3\alpha^2 - 4\alpha - 5 = 0$$

$$P(\beta) = \beta^3 - 3\beta^2 - 4\beta - 5 = 0$$

$$P(\gamma) = \gamma^3 - 3\gamma^2 - 4\gamma - 5 = 0$$

$$\alpha^3 + \beta^3 + \gamma^3 = 3(\alpha^2 + \beta^2 + \gamma^2) + 4(\alpha + \beta + \gamma) + 15$$

$$\text{But } \Sigma \alpha^2 = 17, \quad \Sigma \alpha = 3$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = 3 \times 17 + 4 \times 3 + 15 = 78$$

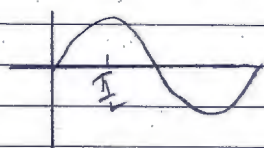
d) $2\sin^3 \theta + 1 = 2\sin^2 \theta + \sin \theta$

$$2\sin^3 \theta - 2\sin^2 \theta - \sin \theta + 1 = 0$$

$$2\sin^2 \theta (\sin \theta - 1) - (\sin \theta - 1) = 0$$

$$(\sin \theta - 1)(2\sin^2 \theta - 1) = 0$$

$$\sin \theta = 1 \quad \text{or} \quad \sin \theta = \pm \frac{1}{\sqrt{2}}$$

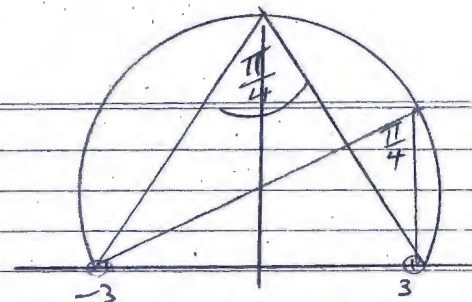


All 4 quadrants

$$\theta = \frac{\pi}{2} \quad \text{or} \quad \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

Question 15

a)



b) i)

$$A = \int_{-1}^1 \frac{1}{x^2+1} - \frac{x^2}{x^2+1} dx$$

$$= 2 \int_0^1 \frac{1-x^2}{x^2+1} dx$$

$$= -2 \int_0^1 \frac{x^2+1-2}{x^2+1} dx$$

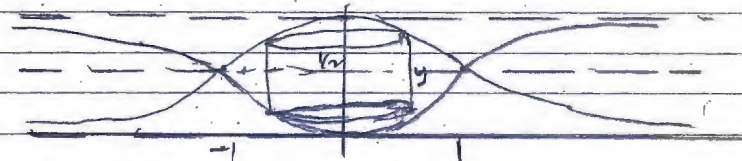
$$= -2 \int_0^1 \left(1 - \frac{2}{x^2+1} \right) dx$$

$$= -2 \left[x - 2 \tan^{-1} x \right]_0^1$$

$$= -2 (1 - 2 \tan^{-1} 1) - 0 + 0$$

$$= -2 (1 - 2 \times \frac{\pi}{4}) = \pi - 2 \quad \text{u}^2$$

ii)



$$V = 2\pi \int_{-1}^1 x y dx$$

where

$$y = \frac{1}{x^2+1} - \frac{x^2}{x^2+1} = \frac{1-x^2}{x^2+1}$$

$$V = 2\pi \int_0^1 x \left\{ \frac{1-x^2}{x^2+1} \right\} dx$$

$$= -2\pi \int_0^1 x \left\{ \frac{x^2+1-2}{x^2+1} \right\} dx$$

$$= -2\pi \int_0^1 x - \frac{2x}{x^2+1} dx$$

$$= -2\pi \left[\frac{x^2}{2} - \ln(x^2+1) \right]_0^1$$

$$= -2\pi \left(\frac{1}{2} - \ln 2 - 0 + \ln 1 \right)$$

$$= \pi (2\ln 2 - 1) u^3$$

g) i) $\int_0^a f(a-x) dx$ Put $y = a-x$
 $dy = -dx$

$$= \int_a^0 f(y) \cdot -dy$$

$$\begin{aligned} x=0, y=a \\ x=a, y=0 \end{aligned}$$

$$= \int_0^a f(y) dy = \int_0^a f(x) dx$$

ii) $\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx = \int_0^\pi \frac{(\pi-x) \sin(\pi-x)}{1+\cos^2(\pi-x)} dx$

$$= \int_0^\pi \frac{(\pi-x) \sin x}{1+(-\cos x)^2} dx$$

$$= \int_0^\pi \frac{\pi \sin x}{1+\cos^2 x} - \frac{x \sin x}{1+\cos^2 x} dx$$

$$2 \int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx = \int_0^\pi \frac{\pi \sin x}{1+\cos^2 x}$$

$$\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx = \left[-\frac{\pi}{2} \tan^{-1} \cos x \right]_0^\pi$$

$$= -\frac{\pi}{2} \tan^{-1} \cos \pi + \frac{\pi}{2} \tan^{-1} \cos 0$$

$$= -\frac{\pi}{2} \tan^{-1}(-1) + \frac{\pi}{2} \tan^{-1}(1)$$

$$= \frac{\pi}{2} \cdot \frac{\pi}{4} + \frac{\pi}{2} \cdot \frac{\pi}{4}$$

$$= \frac{\pi^2}{4}$$

Alternative 15b)

OR $V = \pi \int_0^{\frac{1}{2}} x_1^2 dy + \pi \int_{\frac{1}{2}}^1 x_2^2 dy$

$$y = \frac{x_1^2}{x_1^2 + 1}$$

$$= 1 - \frac{1}{x_1^2 + 1}$$

$$\frac{1}{x_1^2 + 1} = 1 - y$$

$$\frac{1}{1 - y} = x_1^2 + 1$$

$$\frac{1}{1 - y} - 1 = x_1^2$$

$$y = \frac{1}{x_2^2 + 1}$$

$$x_2^2 + 1 = \frac{1}{y}$$

$$x_2^2 = \frac{1}{y} - 1$$

$$\therefore V = \pi \int_0^{\frac{1}{2}} \left(\frac{1}{1 - y} - 1 \right) dy + \pi \int_{\frac{1}{2}}^1 \left(\frac{1}{y} - 1 \right) dy$$

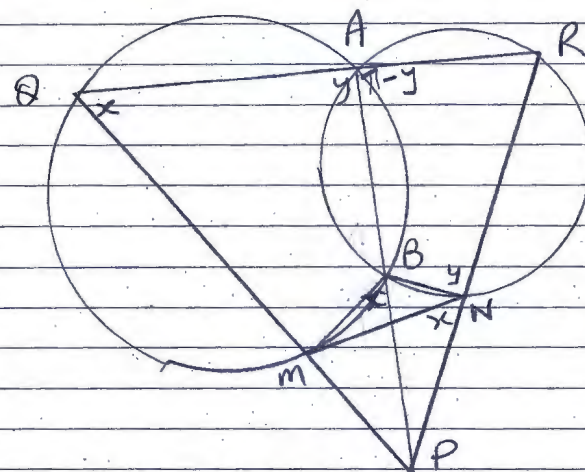
$$V = \pi \left[-\ln(1 - y) - y \right]_0^{\frac{1}{2}} + \pi \left[\ln y - y \right]_{\frac{1}{2}}^1$$

$$= \pi \left[-\ln \frac{1}{2} - \frac{1}{2} + \ln 1 + 0 \right] + \pi \left[\ln 1 - 1 - \ln \frac{1}{2} + \frac{1}{2} \right]$$

$$= \pi \left\{ \ln 2 - \frac{1}{2} - 1 + \ln 2 + \frac{1}{2} \right\}$$

$$= \pi (2 \ln 2 - 1) \quad u^3$$

15
d)



- i)
 $x = \angle MBP = \angle AQM$ Exterior $\angle$ of cyclic quad $BMQA$
 $y = \angle PMB = \angle QAB$ Exterior $\angle$ of cyclic quad $BMQA$
 $\angle RAP = \pi - y$ Straight $\angle$ QAR
 $\angle RNB = y = \pi - (\pi - y)$ Opposite $\angle$'s cyclic quad $ARNB$

$$\therefore \angle RNB = \angle BMP$$

But this is exterior $\angle$ of quad $PMBN$
 $\therefore PMBN$ is a cyclic quad

- ii) $\angle MBP = \angle MNP$ Angles at circumference
 Standing on arc MP of cyclic quad $PMBN$

$$\therefore \angle RQM = \angle MNP = x$$

But $\angle MNP$ is exterior angle of
 quad $QRNM$

$\therefore QRNM$ is a cyclic quadrilateral

Q 11

i) Upwards as Positive $\uparrow$ $\downarrow mg$ $\downarrow mkv^2$

Terminal velocity = U

Downwards $\downarrow mg$ $\uparrow mkv^2$
 $mg - mkv^2 = 0$
 $mkv^2 = mg$

$$kv^2 = g$$

$$v^2 = \frac{g}{k}$$

$$U^2 = \frac{g}{k} \Rightarrow k = \frac{g}{U^2}$$

Upwards motion $\downarrow mg$ $\downarrow mkv^2$

$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -g - kv^2$$

$$\ddot{x} = -g - \frac{g}{U^2} v^2$$

$$\ddot{x} = -\frac{g}{U^2} (U^2 + v^2)$$

ii) $\frac{dv}{dt} = -\frac{g}{U^2} (U^2 + v^2)$

$$\frac{dt}{dv} = -\frac{U^2}{g} \frac{1}{U^2 + v^2}$$

$$t = -\frac{U^2}{g} \cdot \frac{1}{U} \tan^{-1} \frac{v}{U} + C$$

$t=0, v=U \quad 0 = -\frac{U}{g} \tan^{-1} 1 + C$
 $C = \frac{U}{g} \cdot \frac{\pi}{4}$

$$\therefore t = \frac{U}{g} \left\{ \frac{\pi}{4} - \tan^{-1} \frac{v}{U} \right\}$$

$$\frac{gt}{U} = \frac{\pi}{4} - \tan^{-1} \frac{v}{U}$$

$$\tan^{-1} \frac{v}{U} = \frac{\pi}{4} - \frac{gt}{U}$$

$$\frac{v}{U} = \tan \left(\frac{\pi}{4} - \frac{gt}{U} \right)$$

iii) $v = U \tan \left(\frac{\pi}{4} - \frac{gt}{U} \right)$

$$\frac{dx}{dt} = U \tan \left(\frac{\pi}{4} - \frac{gt}{U} \right)$$

$$= U \frac{\sin \left(\frac{\pi}{4} - \frac{gt}{U} \right)}{\cos \left(\frac{\pi}{4} - \frac{gt}{U} \right)}$$

$$x = \frac{U^2}{g} \ln \cos \left(\frac{\pi}{4} - \frac{gt}{U} \right) + C_2$$

$$\therefore \frac{x}{U} = \frac{U}{g} \ln \cos \left(\frac{\pi}{4} - \frac{gt}{U} \right) + C$$

$$\left(\text{as } \frac{d}{dt} \cos \left(\frac{\pi}{4} - \frac{gt}{U} \right) = -\frac{U}{g} \sin \left(\frac{\pi}{4} - \frac{gt}{U} \right) \right)$$

$t=0, x=0 \quad 0 = \frac{U^2}{g} \ln \cos \frac{\pi}{4} + C_2$

$$C_2 = -\frac{U^2}{g} \ln \frac{1}{\sqrt{2}} = \frac{U^2}{g} \ln \sqrt{2}$$

$$\frac{x}{U} = \frac{U}{g} \ln \cos \left(\frac{\pi}{4} - \frac{gt}{U} \right) + \frac{U}{g} \ln \sqrt{2}$$

$$\frac{x}{U} = \frac{U}{g} \ln \cos \sqrt{2} \left(\frac{\pi}{4} - \frac{gt}{U} \right)$$

iv) $t = T$ max height $v = 0$

$$H = \frac{U^2}{g} \ln \sqrt{2} \cos \left(\frac{\pi}{4} - \frac{gT}{U} \right)$$

$$\frac{gT}{U} = \frac{\pi}{4}$$

$$H = \frac{U^2}{g} \ln \sqrt{2} \cos 0$$

$$H = \frac{U^2}{g} \ln \sqrt{2} \quad \text{or} \quad \frac{H}{U} = \frac{U}{2g} \ln 2$$

$$t = \frac{1}{2}T \Rightarrow \frac{g}{U} \left(\frac{1}{2}T \right) = \frac{\pi}{8}$$

$$t = \frac{1}{2}T \Rightarrow \frac{x}{U} = \frac{U}{g} \ln \sqrt{2} \cos \frac{\pi}{8}$$

$$= \frac{U}{2g} \ln 2 \cos^2 \frac{\pi}{8}$$

$$= \frac{U}{2g} \ln \left(1 + \cos \frac{\pi}{4} \right)$$

$$= \frac{U}{2g} \ln \left(1 + \frac{1}{\sqrt{2}} \right)$$

$$t = T \Rightarrow \frac{H}{U} = \frac{U}{g} \ln \sqrt{2} = \frac{U}{2g} \ln 2$$

$$t = \frac{1}{2}T \Rightarrow \frac{x}{H} = \frac{\ln \left(1 + \frac{1}{\sqrt{2}} \right)}{\ln 2}$$

$$\approx 0.77 \quad \text{ie } 77\%$$

of max ht during 1st half of ascent time

Q16(cont)

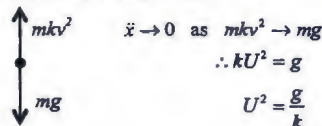
b. Outcomes assessed: E5

Marking Guidelines

Criteria	Marks
i • considers the forces on a falling particle to deduce the value of the square of U	1
• considers the forces on a rising particle to deduce its equation of motion in the form required	1
ii • finds t as a function of v by integration	1
• rearranges to find an expression for v as a function of t	1
iii • integrates with respect to t finding the primitive function	1
• evaluates constant and simplifies to obtain required expression for x as a function of t	1
iv • finds an expressions for T , and hence H , in terms of g and U	1
• finds an expression for x in terms of g and U when half the ascent time has elapsed	1
• calculates the percentage of the maximum height attained at this time	1

Answer

i. Forces on a falling particle



$$ii) \quad \frac{dv}{dt} = -\frac{g}{U^2} (U^2 + v^2)$$

$$\frac{dt}{dv} = -\frac{U}{g} \cdot \frac{U}{U^2 + v^2}$$

$$-\frac{g}{U} t = \tan^{-1} \left(\frac{v}{U} \right) + c$$

$$\left. \begin{matrix} t=0 \\ v=U \end{matrix} \right\} \Rightarrow \left. \begin{matrix} 0 = \tan^{-1} 1 + c \\ 0 = \frac{\pi}{4} + c \end{matrix} \right\}$$

$$\therefore \frac{g}{U} t = \frac{\pi}{4} - \tan^{-1} \left(\frac{v}{U} \right)$$

$$\tan^{-1} \left(\frac{v}{U} \right) = \frac{\pi}{4} - \frac{g}{U} t$$

$$\therefore \frac{v}{U} = \tan \left(\frac{\pi}{4} - \frac{g}{U} t \right)$$

$$ii) \quad t = T \Rightarrow v = 0 \quad \therefore \frac{g}{U} T = \frac{\pi}{4} \quad \text{and} \quad \frac{g}{U} \left(\frac{1}{2} T \right) = \frac{\pi}{8}$$

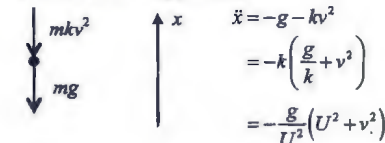
$$\therefore t = \frac{1}{2} T \Rightarrow \frac{x}{U} = \frac{U}{g} \log_e \left\{ \sqrt{2} \cos \frac{\pi}{8} \right\}$$

$$= \frac{U}{2g} \log_e \left(2 \cos^2 \frac{\pi}{8} \right)$$

$$= \frac{U}{2g} \log_e \left(1 + \cos \frac{\pi}{4} \right)$$

$$= \frac{U}{2g} \log_e \left(1 + \frac{1}{\sqrt{2}} \right)$$

Forces on a rising particle



$$iii) \quad \text{Hence} \quad \frac{1}{U} \frac{dx}{dt} = \tan \left(\frac{\pi}{4} - \frac{g}{U} t \right)$$

$$\frac{1}{U} \frac{dx}{dt} = \frac{\sin \left(\frac{\pi}{4} - \frac{g}{U} t \right)}{\cos \left(\frac{\pi}{4} - \frac{g}{U} t \right)}$$

$$\frac{x}{U} = \frac{U}{g} \log_e \left\{ \cos \left(\frac{\pi}{4} - \frac{g}{U} t \right) \right\} + c_1$$

$$\left. \begin{matrix} t=0 \\ x=0 \end{matrix} \right\} \Rightarrow \left. \begin{matrix} 0 = \frac{U}{g} \log_e \left(\cos \frac{\pi}{4} \right) + c_1 \\ 0 = \frac{U}{g} \log_e \frac{1}{\sqrt{2}} + c_1 \end{matrix} \right\}$$

$$\frac{x}{U} = \frac{U}{g} \log_e \left\{ \sqrt{2} \cos \left(\frac{\pi}{4} - \frac{g}{U} t \right) \right\}$$

$$t = T \Rightarrow \frac{x}{U} = \frac{U}{g} \log_e \sqrt{2} \quad \therefore \frac{H}{U} = \frac{U}{2g} \log_e 2$$

$$\therefore t = \frac{1}{2} T \Rightarrow \frac{x}{H} = \frac{\log_e \left(1 + \frac{1}{\sqrt{2}} \right)}{\log_e 2} \approx 0.77$$

Hence particle gains 77% of its maximum height during the first half of its ascent time.

1/6 b)

Tangent at P is $x + p^2 y = 2cp$ — (1)

Tangent at Q is $x + q^2 y = 2cq$ — (2)

① - ② $(p^2 - q^2)y = 2c(p - q)$

$p \neq q \therefore (p+q)y = 2c$
 $y = \frac{2c}{p+q}$

Sub into ①

$x + p^2 \frac{2c}{p+q} = 2cp$

$(p+q)x + 2cp^2 = 2cp^2 + 2cpq$
 $x = \frac{2cpq}{p+q}$

Tangents meet at R (x_0, y_0)

$y_0 = \frac{2c}{p+q}$ or $p+q = \frac{2c}{y_0}$

and $x_0 = \frac{2cpq}{2c/y_0} = pq y_0$

$\therefore pq = \frac{x_0}{y_0}$

ii) Distance PQ is d

$d^2 = (cp - cq)^2 + \left(\frac{c}{p} - \frac{c}{q}\right)^2$
 $= c^2(p-q)^2 + c^2\left(\frac{q-p}{pq}\right)^2$

$d^2 = c^2(p-q)^2 \left\{1 + \frac{1}{p^2 q^2}\right\}$

iii) Note $(p-q)^2 = (p+q)^2 - 4pq$

$= \left(\frac{2c}{y_0}\right)^2 - \frac{4x_0}{y_0}$

Substitute into d^2

$d^2 = c^2 \left\{ \frac{4c^2}{y_0^2} - \frac{4x_0}{y_0} \right\} \left\{ 1 + \frac{y_0^2}{x_0^2} \right\}$

Multiply by $x_0^2 y_0^2$

$x_0^2 y_0^2 d^2 = 4c^2 (c^2 - x_0 y_0) (x_0^2 + y_0^2)$

(x_0, y_0) lies on curve with equation

$x^2 y^2 d^2 = 4c^2 (c^2 - xy) (x^2 + y^2)$

$\therefore$ this is the locus of R as c and d are constants.

